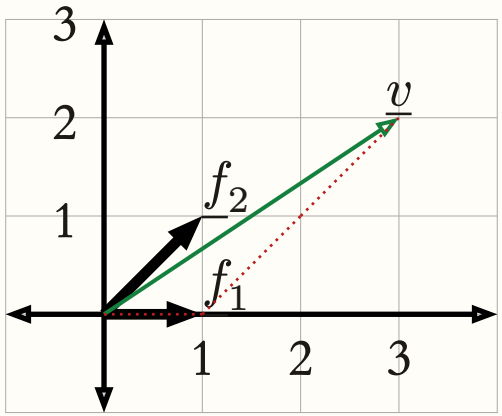
Linear Algebra III: transformations

Week 4 — Algorithmic Approaches to Mathematics

Dhruva Raman

Recap: matrices convert co-ordinates

Two bases

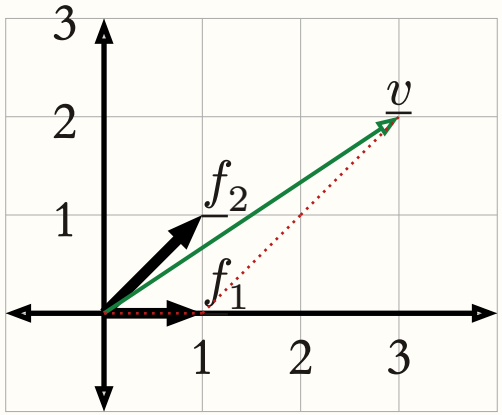


$$\underline{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \underline{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \underline{f}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \underline{f}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\underline{v} = 3\underline{e}_1 + 2\underline{e}_2$$

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Two bases



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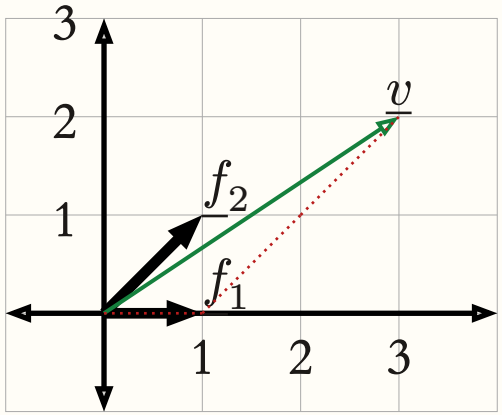
$$\underline{v} = 1\underline{f}_1 + 2\underline{f}_2$$

Converting f-coordinates into e-coordinates

$$\begin{bmatrix}1 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}3\\2\end{bmatrix}$$

The columns of the matrix are the new basis vectors

Two bases



$$\underline{e}_1 = \begin{bmatrix}1\\0\end{bmatrix} \quad \underline{e}_2 = \begin{bmatrix}0\\1\end{bmatrix} \quad \underline{f}_1 = \begin{bmatrix}1\\0\end{bmatrix} \quad \underline{f}_2 = \begin{bmatrix}1\\1\end{bmatrix}$$

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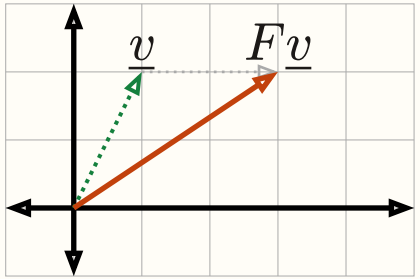
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The columns of the matrix are the new basis vectors

Reverse (hwk): $\begin{bmatrix}1 & -1\\0 & 1\end{bmatrix}\begin{bmatrix}3\\2\end{bmatrix} = \begin{bmatrix}1\\2\end{bmatrix}$

Yes. Backwards conversion is the inverse matrix

The duality of matrices

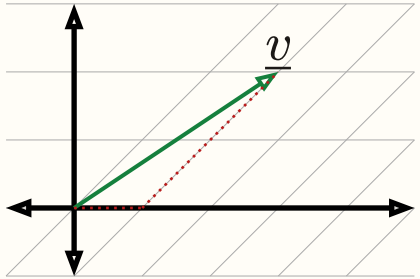


Reading 1: transform the vector

Grid stays. F shears space:

$$F\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}3\\2\end{bmatrix}$$

the vector moves



Reading 2: change the co-ordinates[1]

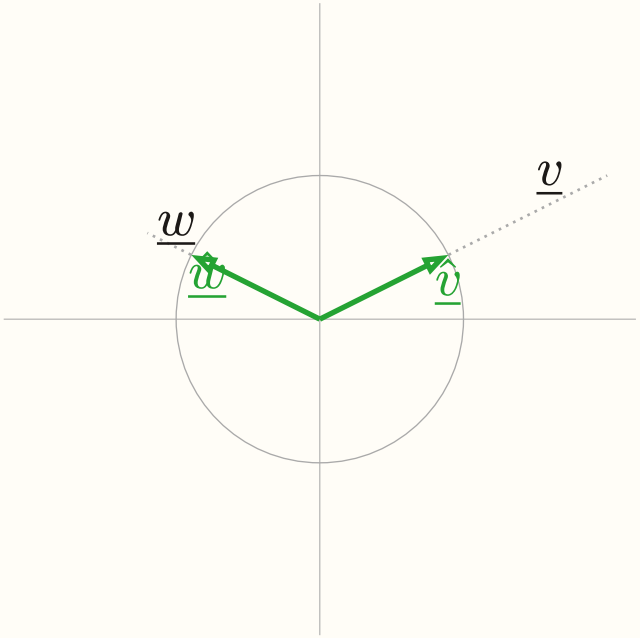
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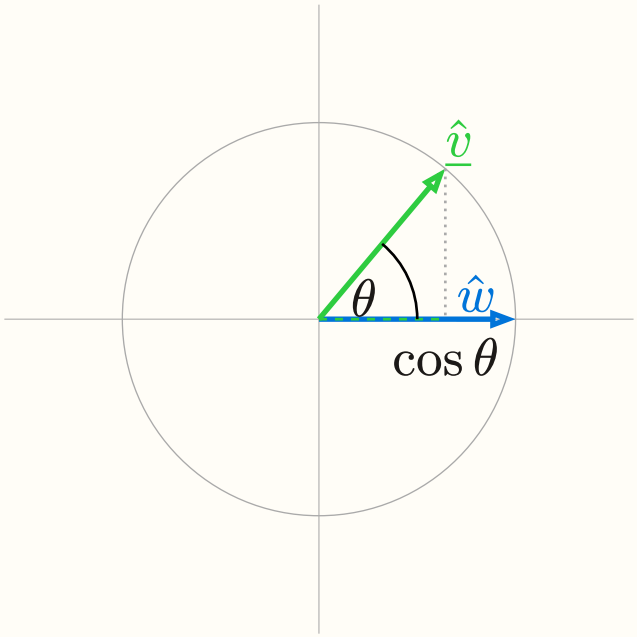
$(1;2)$ are f-coordinates of the vector at $(3;2)$

[1] great video here

Correlation, again (recap)



Shrink onto the unit circle…



…then dot products are cosines

Dot product practice

Useful identity

$$\underline{v}^T \underline{v} = \|\underline{v}\|_2^2$$

e.g. $\underline{v} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$

$$\underline{v}^T \underline{v} = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \sum_i v_i^2$$

Question: Geometrical interpretation of $\underline{v}^T \underline{w} = \underline{0}$

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Question: Simplify $\|A\underline{v}\|_2^2$

$$\|A\underline{v}\|_2^2 = (A\underline{v})^T (A\underline{v}) = \underline{v}^T A^T A \underline{v}$$

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Describe A geometrically if

$\langle A\underline{v}, A\underline{w} \rangle = \langle \underline{v}, \underline{w} \rangle \quad \forall \underline{v}, \underline{w}$

Dot product practice

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Rotation: preserves angles. $A^T A = \mathbb{I}$

All about linearity

What is linearity?

Definition: linear map[1] between vector spaces

$T : U \to V$ is a linear map if

$$T(\underline{v} + \underline{w}) = T(\underline{v}) + T(\underline{w}) \quad \forall \underline{v}, \underline{w} \in U$$

$$T(c\underline{v}) = cT(\underline{v}) \quad \forall c \in K, \underline{v} \in U$$

[1] map = function. why map? convention I guess

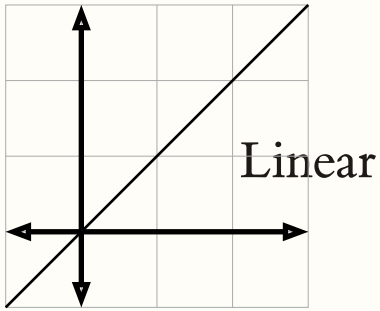
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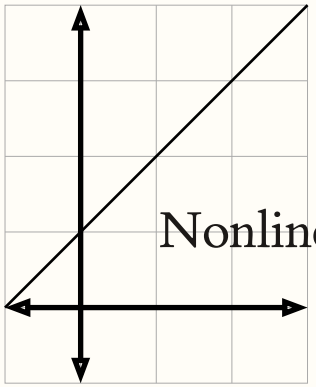
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- Why?

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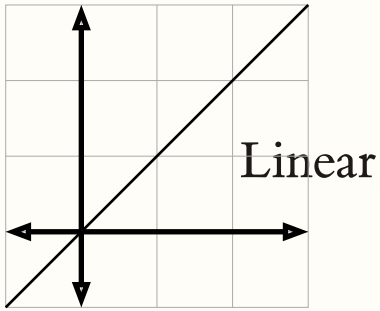
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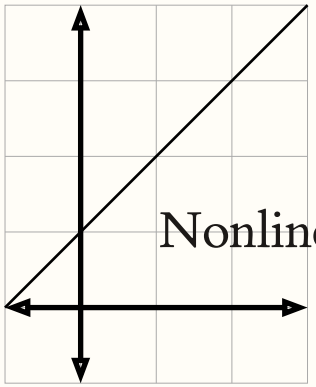
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- Why?
- P.S second map called affine (linear + constant)

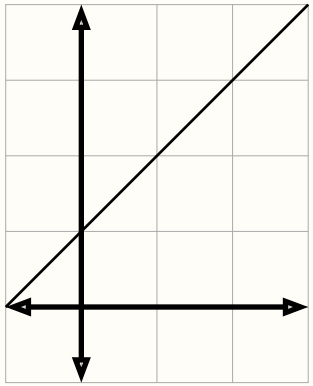
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Not to be confused: affine maps

Affine = linear + shift

$$f(x) = Ax + b$$

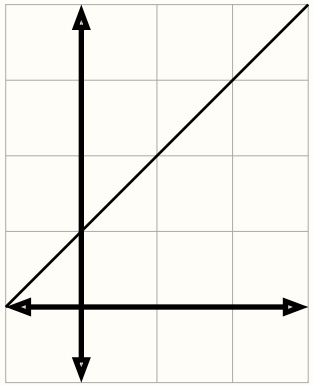
Example: $f(x) = x + 1$



$$f(c0) = f(0) \neq cf(0) = c$$

Not to be confused: affine maps

Example: $f(x) = x + 1$



$f(c0) = f(0) \neq cf(0) = c$

Affine = linear + shift

$$f(x) = Ax + b$$

- e.g. every neural-network layer, pre-activation:

$$\underline{y} = \sigma(W\underline{x} + \underline{b})$$

Optional fast-forward

- Chaining affine functions gives affine.
- Hence neural networks need nonlinearities (σ: ReLu, tanh, etc)

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Scrunched definition:[1]

$$T(c\underline{v} + d\underline{w}) = cT(\underline{v}) + dT(\underline{w})$$
$$\forall c, d \in K; \quad \underline{v}, \underline{w} \in U$$

Exercise

- Verify linearity of $f(x) = 4x$

[1] Equivalent to previous but in one go

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Exercise

- Verify linearity of $f(x) = 4x$
- Verify linearity of $f : \mathbb{R}^m \to \mathbb{R}^n$, $f(\underline{x}) = A\underline{x}$

[1] Equivalent to previous but in one go

Linearity practice

$$T(c\underline{v} + d\underline{w}) = cT(\underline{v}) + dT(\underline{w})$$

Linear map?

$$f(x, y) = (y, x)$$

[1] Note switching between x_i and x, y notation

[2] Mathematicians forced to write multiplication use $\cdot$ instead of $\times$

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Yes[1]: $f(\underline{x}) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

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$$f(x) = x^2$$

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Linearity practice

$$T(c\underline{v} + d\underline{w}) = cT(\underline{v}) + dT(\underline{w})$$

Linear map?

Rotate by 90°

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$$f(x) = x^2$$

No[2]: $f(2 \cdot 1) = 4 \neq 2f(1)$

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No[2]: $f(2 \cdot 1) = 4 \neq 2f(1)$

Rotate by 90°

Intuition says yes but also $f(\underline{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \underline{x}$

$$f(x, y) = (x, 0)$$

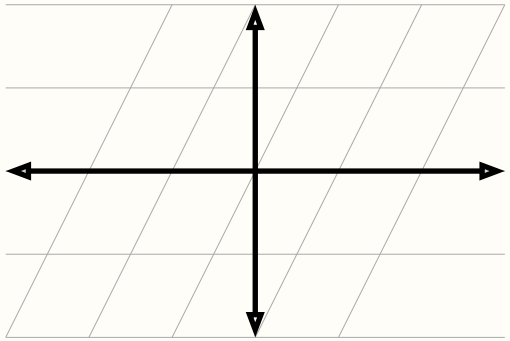
Yes: projection (see cheatsheet)

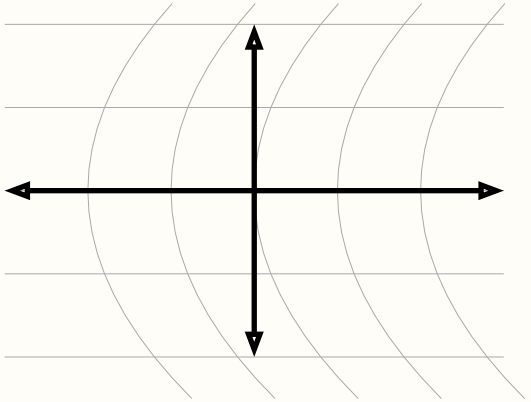
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Linearity, geometrically

$$T(\underline{v}) = \underline{v} \text{ in new basis}$$





Linear

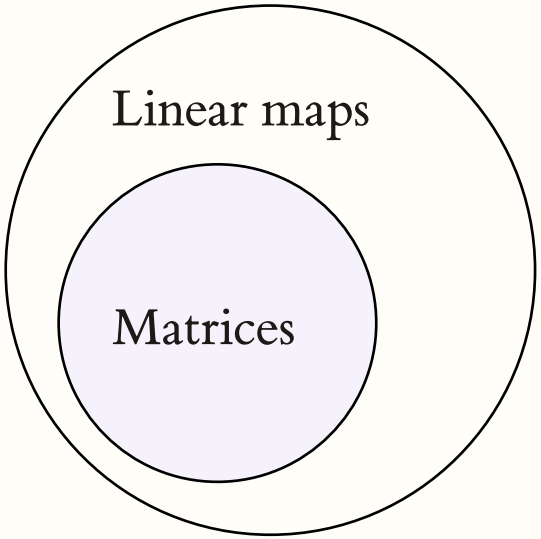
- New co-ordinate grid has straight lines.
- Origin fixed ($T(\underline{0}) = \underline{0}$)

Nonlinear

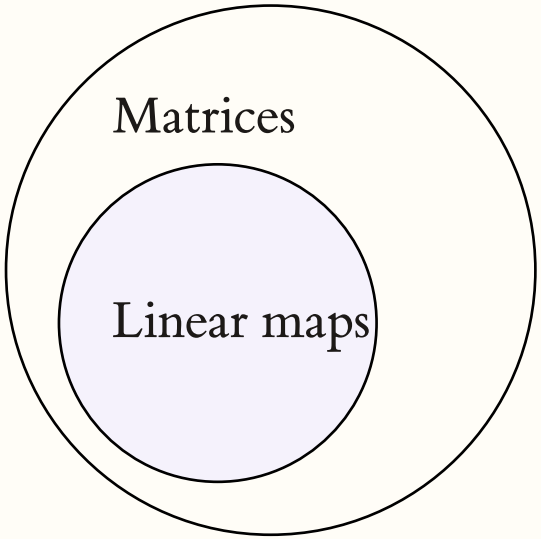
- Anything else
- E.G. new co-ordinate grid has curved lines.

Preview: Linear maps must be matrices

Before



Goal



Composing linear maps

=

Multiplying matrices

Common maths technique

$A \Rightarrow B$ and $B \Rightarrow A$ means that $A \Leftrightarrow B$!

What's the deal with linear maps anyway?

The situation

Vector space V with basis $\{\underline{e}_1, \underline{e}_2, \ldots, \underline{e}_n\}$

Given n pieces of data only:

$$T(\underline{e}_i) := \underline{f}_i$$

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Question

Can you tell me how T transforms any vector in V?

Example: $\mathbb{R}^2 \to \mathbb{R}^3$

$$T(\underline{e}_1) = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \quad T(\underline{e}_2) = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

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Extrapolated entire linear map from two dots!

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Correspondence between matrices and linear maps

$$M = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -3 & -2 \end{bmatrix}$$

Columns specify what each basis in the domain maps to

Co-ordinates of output? Some basis of image

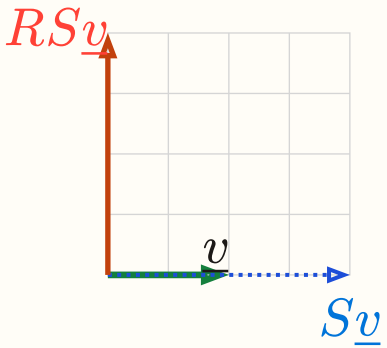
$$T(3\underline{e}_1 + \underline{e}_2 + 2\underline{e}_3) = M \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \end{bmatrix}$$

Composing linear maps

Stretch x by 2, then rotate anticlockwise by 90°:

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \qquad R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Do S, then R



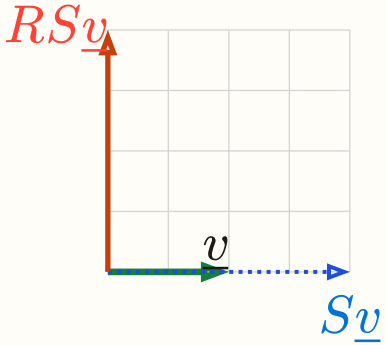
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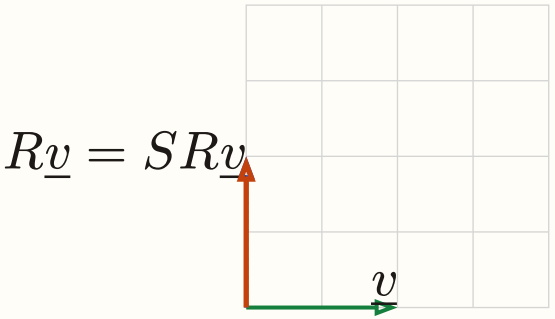
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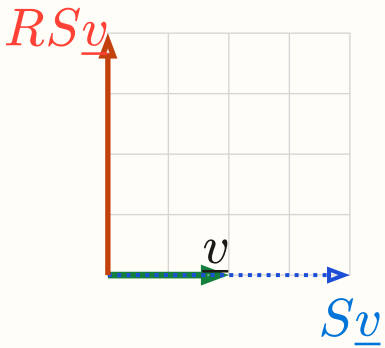
Do R, then S

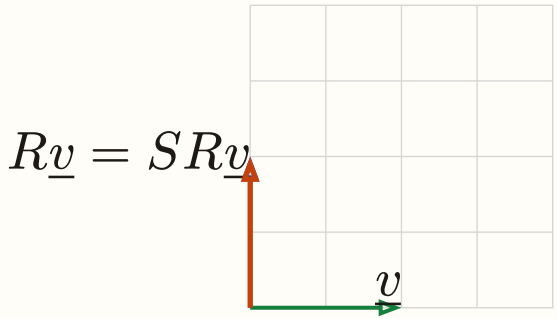


$$SR\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Composing linear maps

A matrix product is a composition, read right to left: AB means "do B, then A"





Why must $(AB)^{-1} = B^{-1}A^{-1}$ and $(AB)^T = B^T A^T$?[1]

[1] Think about taking on/off socks and shoes

Back to groceries! (recap)

Old

$$\begin{bmatrix}\text{prices} \rightarrow \\ \text{weights} \rightarrow\end{bmatrix} \left[\begin{array}{cc}1.50 & 0.75 \\ \hdashline 0.5 & 0.25\end{array}\right] \begin{bmatrix}🍏 \\ 🍌\end{bmatrix}$$

$$\left[\begin{array}{cc}1.50 & 0.75 \\ \hdashline 0.5 & 0.25\end{array}\right] \begin{bmatrix}3 \\ 2\end{bmatrix} = \begin{bmatrix}6 \\ 2\end{bmatrix} \left(\begin{bmatrix}£ \\ \text{kg}\end{bmatrix} \right)$$

Rows on left ask questions

- price/weight of all items

Columns on right answer

- Per-basket

Back to groceries! (recap)

Old

$$\begin{bmatrix}\text{prices} \rightarrow \\ \text{weights} \rightarrow\end{bmatrix}\ \left[\begin{array}{cc}1.50 & 0.75 \\ \hdashline 0.5 & 0.25\end{array}\right]\ \begin{bmatrix}🍏 \\ 🍌\end{bmatrix}$$

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- price/weight of all items

Columns on right answer

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New

$$\left[\begin{array}{c|c}1.50 & 0.75 \\ 0.5 & 0.25\end{array}\right]\begin{bmatrix}3 \\ 2\end{bmatrix} = 3\begin{bmatrix}1.50 \\ 0.5\end{bmatrix} + 2\begin{bmatrix}0.75 \\ 0.25\end{bmatrix}$$

Rows on right ask questions

- How many of each fruit?

Columns on left answer

- All per-fruit data

$$A = \begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix}$$

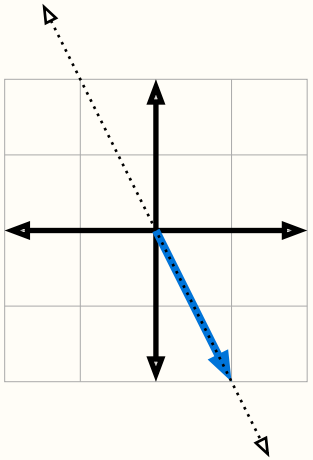
Kernel of A

Set of baskets costing £0, weighing 0lb:

$$\ker(A) = \text{span}\left\{\begin{bmatrix} 1 \\ -2 \end{bmatrix}\right\}$$

One-dimensional vector space

i.e. a line

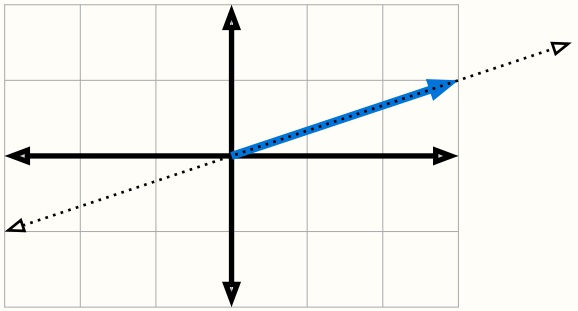


Kernel of a matrix

$$\ker(A) = \{\underline{x} : A\underline{x} = \underline{0}\}$$

Kernel and nullity

$$A = \begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix}$$



Reachable (cost, weight) pairs:

$$\mathfrak{I}(A) = \text{span}\left\{\begin{bmatrix} 3 \\ 1 \end{bmatrix}\right\}$$

One-dimensional vector space

i.e. a line

Image of a matrix

$$\mathfrak{I}(A) = \left\{\underline{y} : \underline{y} = A\underline{x}\right\}$$

$$\text{rank}(A) = \dim(\mathfrak{I}(A))$$

Rank-nullity theorem

Definition

Let $A \in \mathbb{R}^{m \times n}$. Then

$$\operatorname{rank}(A) + \dim(\ker(A)) = n$$

Proof

Difficult but possible. Widely available online.

Takeaway

Input dimensions survive (rank) or die (nullity)

Rank-nullity theorem

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Example

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \\ 4 & 8 & 0 \end{bmatrix}$$

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$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 5 \\ 4 & 8 & 0 \end{bmatrix}$$

$$A\underline{x} = (x_1 + 2x_2) \begin{bmatrix} 1 \\ 4 \\ 8 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$$

$\operatorname{rank}(A)$? $\dim(\ker(A))$?

$$\ker(A) = \operatorname{span}\left\{ \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \right\}$$

$$A\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}11\\25\\20\end{bmatrix}$$

Solution set

$$\left\{\underline{x} : A\underline{x} = \begin{bmatrix}11\\25\\20\end{bmatrix}\right\}?$$

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Solution set

$$\left\{ \underline{x} : A\underline{x} = \begin{bmatrix} 11 \\ 25 \\ 20 \end{bmatrix} \right\}?$$

$$= \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \underline{v} : v \in \ker(A) \right\}$$

Example

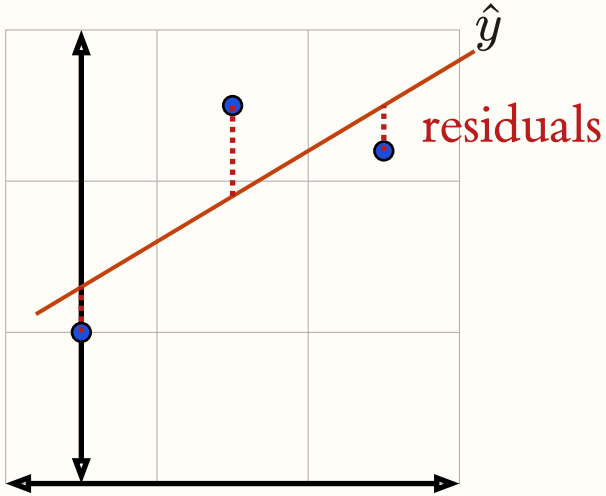
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A regression teaser



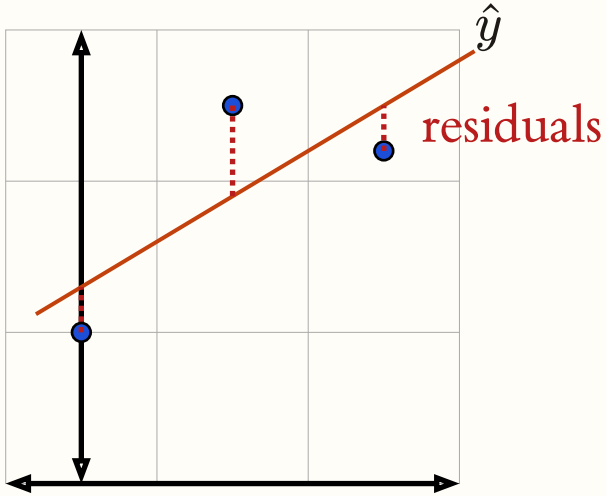
Fit $\underline{y} \approx a + bx$ to three (or more) points

As a matrix equation

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \approx \begin{bmatrix} 1 \\ 2.5 \\ 2.2 \\ \vdots \end{bmatrix}$$

Rank? Nullity?

A regression teaser



Fit $\underline{y} \approx a + bx$ to three (or more) points

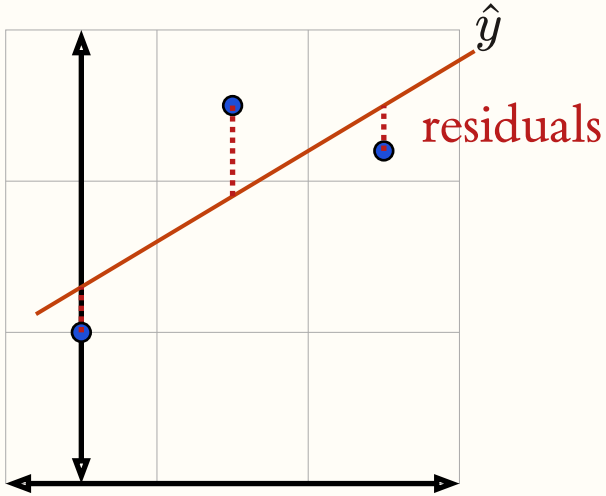
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Rank? Nullity? 2 and 0

No exact solution: $\underline{y}$ is not in $\mathfrak{I}(M)$

A regression teaser



Fit $\underline{y} \approx a + bx$ to three (or more) points

Optimisation

Make $M \underbrace{\begin{bmatrix} a \\ b \end{bmatrix}}_{\tilde{x}}$ as close to $\underline{y}$ as possible:

$$\tilde{x} = \text{argmin}_x \left\| Mx - \underline{y} \right\|_2^2$$

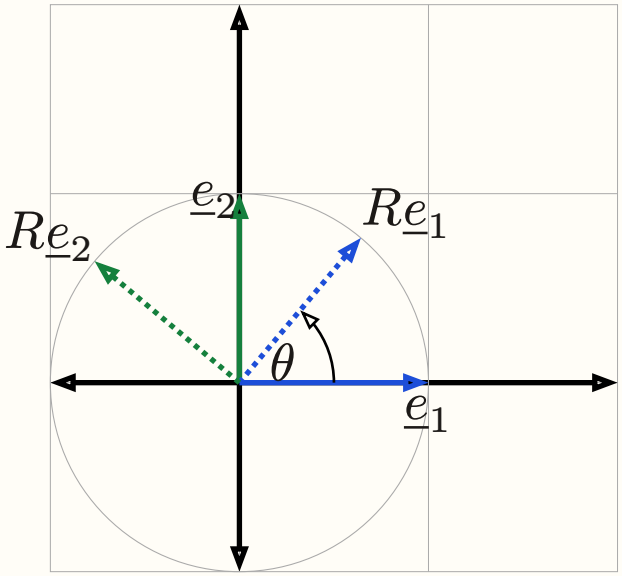
i.e.

$$\min_{a,b} \left\| \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ \vdots & \vdots \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} - \begin{bmatrix} 1 \\ 2.5 \\ 2.2 \\ \vdots \end{bmatrix} \right\|_2^2$$

Maps, geometrically

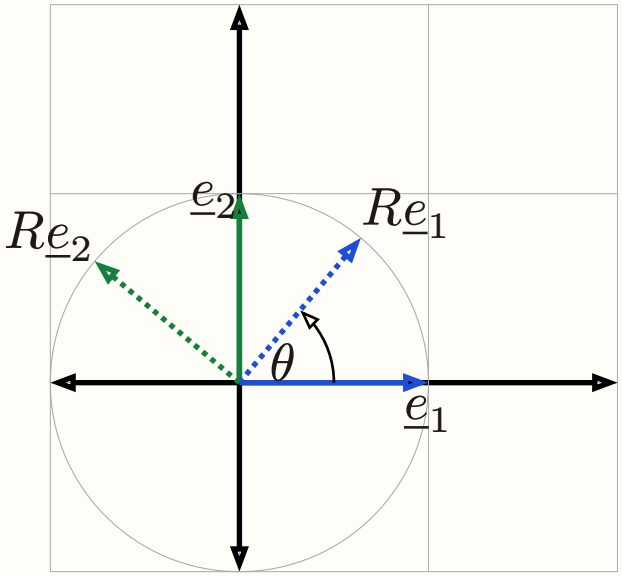
rotation, scaling, and directions that survive

Rotation



Can we express this as a matrix?

Rotation

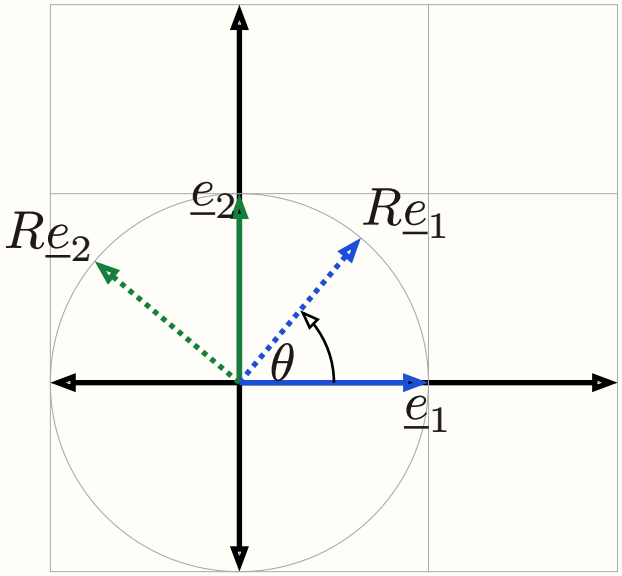


Can we express this as a matrix?

1. **Where do basis vectors go?**

$$\underline{e}_1 \to \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \qquad \underline{e}_2 \to \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

Rotation



Can we express this as a matrix?

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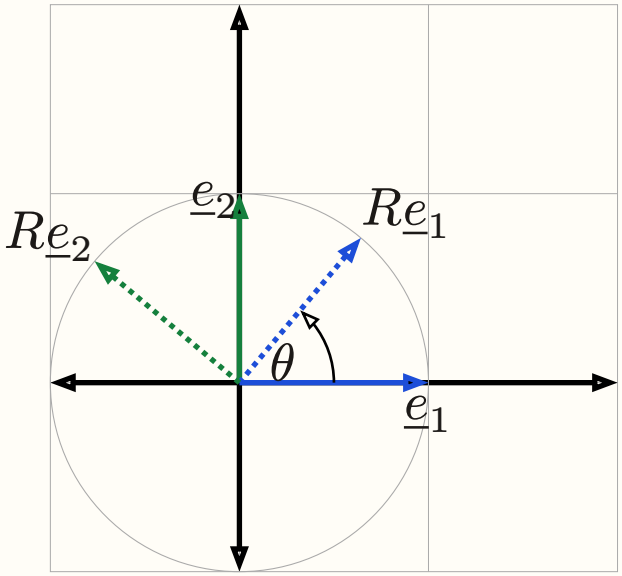
$$\underline{e}_1 \to \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} \qquad \underline{e}_2 \to \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

2. Done!

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Columns = images of the basis

Rotation

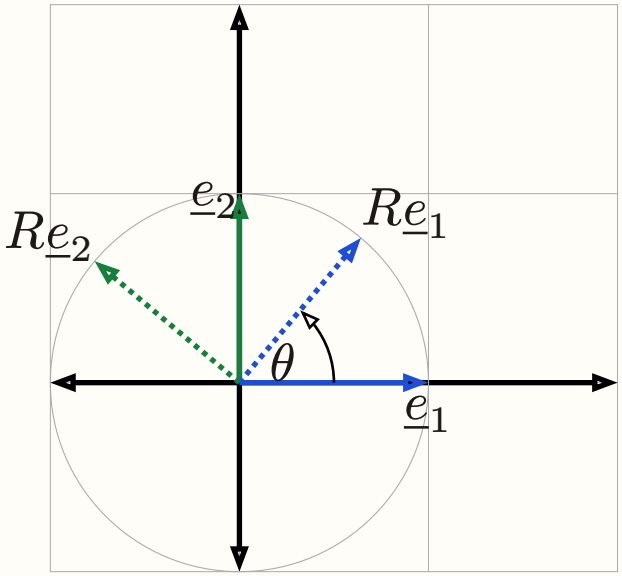


Can we express this as a matrix?

Question

What happens to dot product of two vectors under rotation?

Rotation



Can we express this as a matrix?

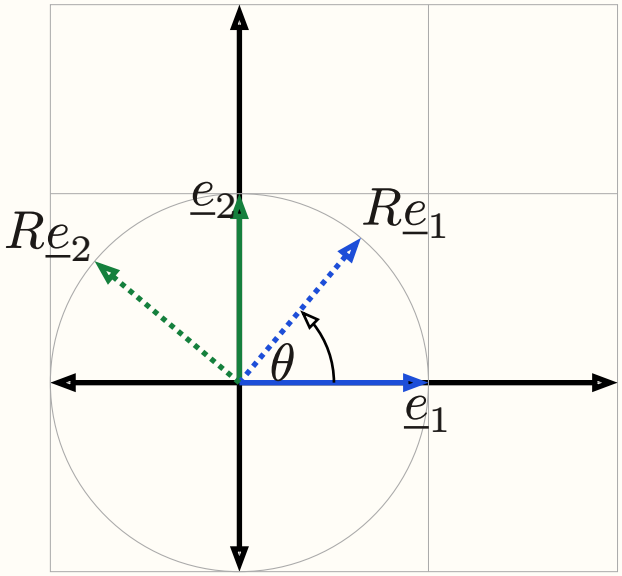
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Nothing! Express this mathematically

$$\underline{v}^T \underline{w} =$$

Rotation



Can we express this as a matrix?

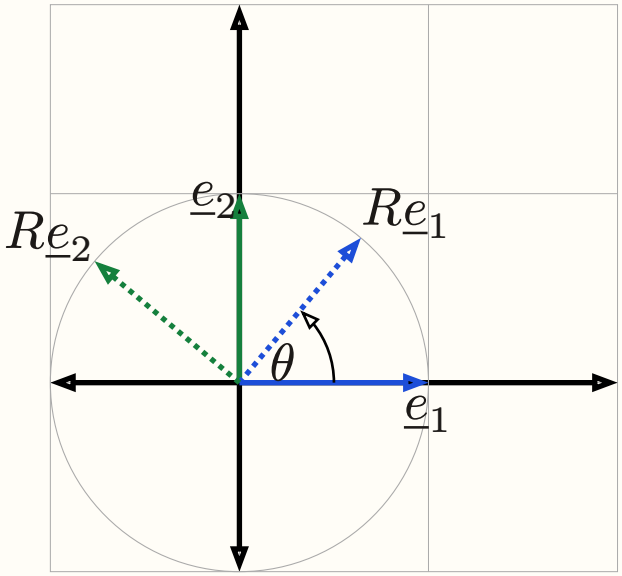
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Can we express this as a matrix?

Question

What happens to dot product of two vectors under rotation?

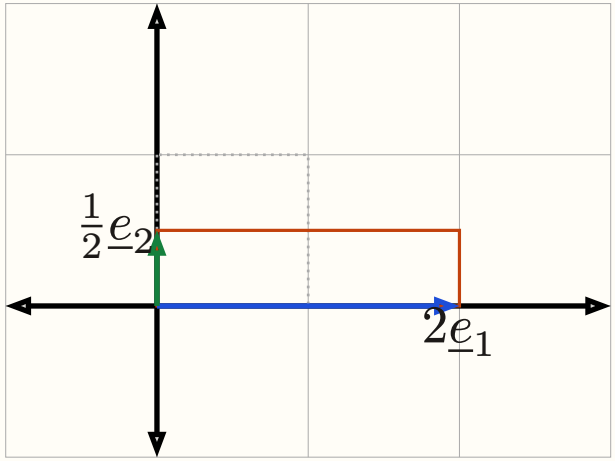
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$$R_\theta^T R_\theta = \mathbb{I}_n$$

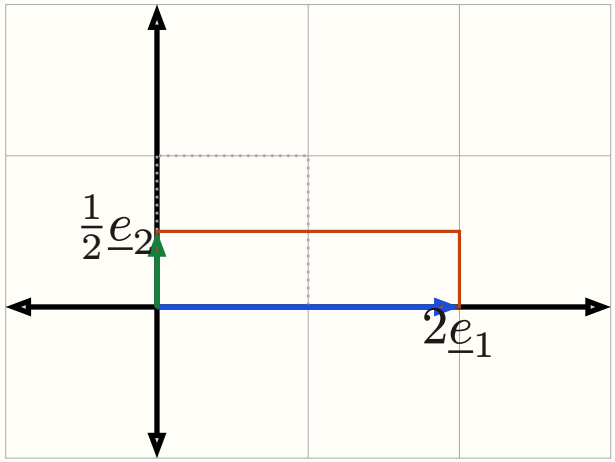
Scaling



$$S = \begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$\underline{e}_1 \to 2\underline{e}_1, \quad \underline{e}_2 \to \frac{1}{2}\underline{e}_2$$

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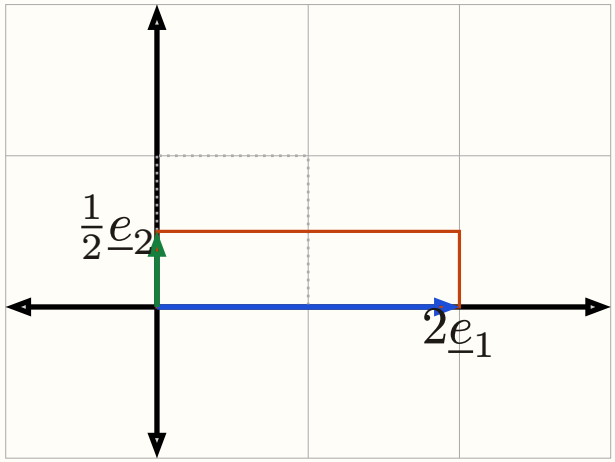
Practice: build matrices for

- 90° rotation
- Reflection in the x-axis

Harder

- is $R_\alpha R_\beta$ a rotation? Implies which trig identities?

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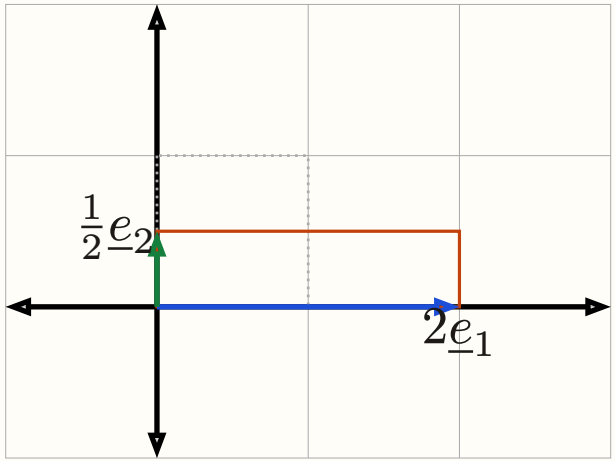
- 90° rotation
- Reflection in the x-axis

Harder

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Knowing how linear map transforms bases tells you all you need to know!

Scaling



$$S = \begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

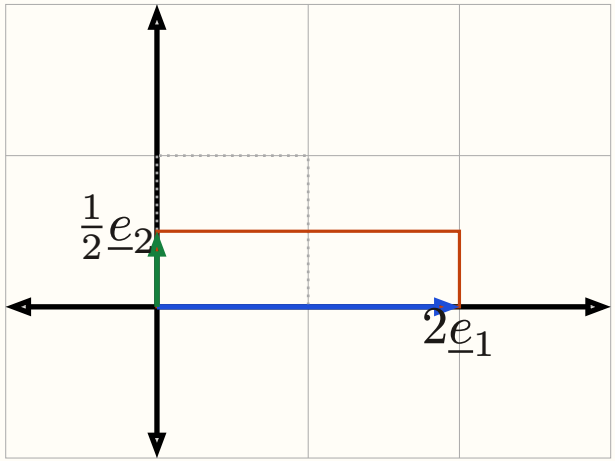
$$\underline{e}_1 \to 2\underline{e}_1, \quad \underline{e}_2 \to \frac{1}{2}\underline{e}_2$$

Scaling didn't rotate co-ordinates

What about the sparsity pattern[1] tells you this?

[1] pattern of 0s vs not 0s

Scaling



$$S = \begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

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Scaling didn't rotate co-ordinates

What about the sparsity pattern[1] tells you this?

Arbitrary matrix:

$$A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

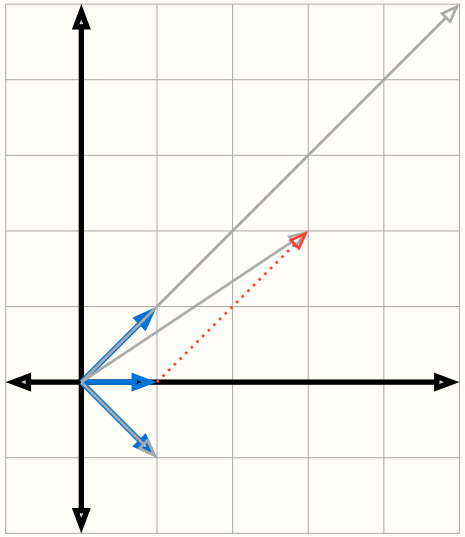
Are there any directions this matrix scales without rotation?

[1] pattern of 0s vs not 0s

Eigenvectors of a (square) matrix

$$\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} \qquad \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$



Purely scaled directions are called eigenvectors

Degree of scaling is the associated eigenvalue

Eigenvector/value pair: $(\underline{v}, \lambda)$[1]

[1] `np.linalg.eig(A)`

Eigenvectors of a (square) matrix

Definition

$v \in \mathbb{R}^n$ is an eigenvector of $A \in \mathbb{R}^{n \times n}$ if

$$A\underline{v} = \lambda \underline{v},$$

for some scalar $\lambda \in K$ called the associated eigenvalue

Eigenvectors/values of…

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}?$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}?$$

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}?$$

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Rotation will only have complex eigenvalues!

Bit of a random definition?

Eigenvectors pop up everywhere

We will see them again

Covariance matrices:
Statistics and Principal Component Analysis

Hessians:
Curvature in gradient descent (week 8)

Dynamical systems:
Eigenvalues decide stability (week 10)

Interlude: variance of a vector

Variance is a measure of spread around the mean

$$\underline{v} = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 8 \end{bmatrix}; \quad \mu(\underline{v})? \text{ (mean)}$$

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Definition

$$\text{var}(\underline{v}) = \left(\frac{1}{n}\right) \sum_i (v_i - \mu(\underline{v}))^2$$

$$= \frac{\|\underline{v} - \mu(\underline{v})\|_2^2}{n}$$

$\text{var}(2\underline{v})$?

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$$= \frac{\|\underline{v} - \mu(\underline{v})\|_2^2}{n}$$

$$\text{var}(2\underline{v})?$$

$$= 4 \text{ var}(\underline{v})$$

$$\text{var}(c\underline{v}) = c^2 \text{ var}(\underline{v}) \quad \forall c \in K, \underline{v} \in V$$

Recap: orthogonal/rotation matrices

A is a rotation

$$\underline{v}^\top \underline{w} = \langle \underline{v}, \underline{w} \rangle = k$$

What is

$$\langle A\underline{v}, A\underline{w} \rangle ?$$

Remember

Dot product $\propto$ angle times magnitude

Recap: orthogonal/rotation matrices

A is a rotation

$$\underline{v}^\top \underline{w} = \langle \underline{v}, \underline{w} \rangle = k$$

What is

$$\langle A\underline{v}, A\underline{w} \rangle ?$$

$$= k!$$

Write $\langle A\underline{v}, A\underline{w} \rangle$ as matrix product?

Remember

Dot product $\propto$ angle times magnitude

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$$= k!$$

Write $\langle A\underline{v}, A\underline{w} \rangle$ as matrix product?

$$= \underline{v}^\top A^\top A \underline{w}$$

Rotation: $A^\top A = \mathbb{I}$!

Remember

Dot product $\propto$ angle times magnitude

Back to netflix!

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	−1	−1
Bob	1	1	−1	−1
Carol	−1	−1	1	1
Eve	1	−1	0	0
Frank	10	10	−10	−10

Meaning of $A^T \underline{v} \in \mathbb{R}^4$?

Back to netflix!

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Meaning of $A^T \underline{v} \in \mathbb{R}^4$?

$$\begin{bmatrix} 1 & 1 & -1 & 1 & 10 \\ 1 & 1 & -1 & -1 & 10 \\ -1 & -1 & 1 & 0 & -10 \\ -1 & -1 & 1 & 0 & -10 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 13 \\ 13 \\ -13 \\ -13 \end{bmatrix}$$

Alice gets two votes, Carol/Eve get none!

- Jury opinion on films (weighted vote)

Back to netflix!

Most informative jury?

- Votes disagree (large spread)

$$\begin{bmatrix} 1 & 1 & -1 & 1 & 10 \\ 1 & 1 & -1 & -1 & 10 \\ -1 & -1 & 1 & 0 & -10 \\ -1 & -1 & 1 & 0 & -10 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 13 \\ 13 \\ -13 \\ -13 \end{bmatrix}$$

Not most informative as Alice/Bob/Frank always agree

Back to netflix!

Most informative jury?

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Disagreement of the jury

$$\text{var}(\underline{u}) = \left\| A^T \underline{u} \right\|_2^2 = \underline{u}^T (AA^T) \underline{u}$$

- Already mean-centered

Back to netflix!

Most informative jury?

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$$\underline{u}_1 = \text{maximise} \quad \underline{u}^T (AA^T) \underline{u} \quad \text{over } \|\underline{u}\|_2 = 1$$

Back to netflix!

Sample covariance matrix:
(person-person alignments)

$$AA^\top = \begin{bmatrix} 4 & 4 & -4 & 0 & 40 \\ 4 & 4 & -4 & 0 & 40 \\ -4 & -4 & 4 & 0 & -40 \\ 0 & 0 & 0 & 2 & 0 \\ 40 & 40 & -40 & 0 & 400 \end{bmatrix}$$

Disagreement of the jury

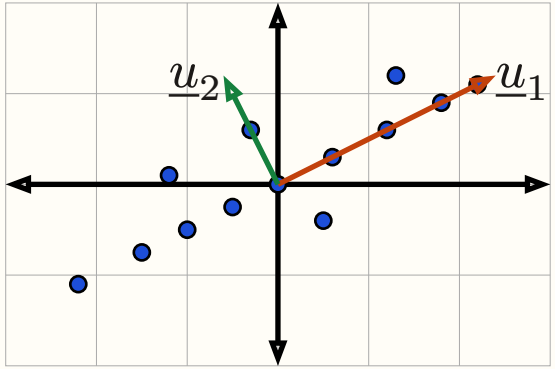
$$\mathrm{var}(\underline{u}) = \left\| A^T \underline{u} \right\|_2^2 = \underline{u}^T (AA^T) \underline{u}$$

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A preview of Principal Component Analysis

1



- Centred data cloud (4D not 2D)
- Each dot analogous to Netflix person
- $\underline{u}_1$: *composite* person that best captures the crowd

$$\underline{u}_1 = \text{maximise} \quad \underline{u}^T(AA^T)\underline{u} \quad \text{over } \|\underline{u}\|_2 = 1$$

PCA changes the basis

Basis 1: maximises variance (jury disagreement)

Basis i: maximises variance and orthogonal to previous bases

[1] PCA computed by "singular value decomposition", coming up…

So I did PCA

Change of basis matrix (rotation):

$$V = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{2}}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{\sqrt{2}}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

Same matrix, new co-ordinates :

$$S = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 20 & 0 & 0 & 0 \end{bmatrix}$$

V creates movie flavours

S : preference to the composite movies

$$A = SV^\top$$

where S is "scores" matrix

People only care about two flavours of movie!

Inspecting the movie rotation matrix

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$$V = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{2}}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{\sqrt{2}}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \quad V^\top V = \mathbb{I}$$

$$A = SV^T$$
$$AV = S$$

Inspecting the movie rotation matrix

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$$V = \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{2}}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{\sqrt{2}}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \qquad V^\top V = \mathbb{I}$$

$$A = SV^T$$
$$AV = S$$

Most engaging composite movie is $V_{\bullet,1}$

Second most is $V_{\bullet,2}$ (excites Eve)

Only two composite movies!

$$S = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 20 & 0 & 0 & 0 \end{bmatrix}$$

Now let's rotate many people into few personalities

$$S = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 20 & 0 & 0 & 0 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 2\sqrt{103} & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$U = \begin{bmatrix} \frac{1}{\sqrt{103}} & 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{101}{\sqrt{20806}} \\ \frac{1}{\sqrt{103}} & 0 & -\frac{1}{\sqrt{2}} & 0 & -\frac{101}{\sqrt{20806}} \\ -\frac{1}{\sqrt{103}} & 0 & 0 & \frac{10}{\sqrt{101}} & -\frac{2}{\sqrt{20806}} \\ 0 & 1 & 0 & 0 & 0 \\ \frac{10}{\sqrt{103}} & 0 & 0 & \frac{1}{\sqrt{101}} & \frac{20}{\sqrt{20806}} \end{bmatrix}$$

U rotates people into personalities

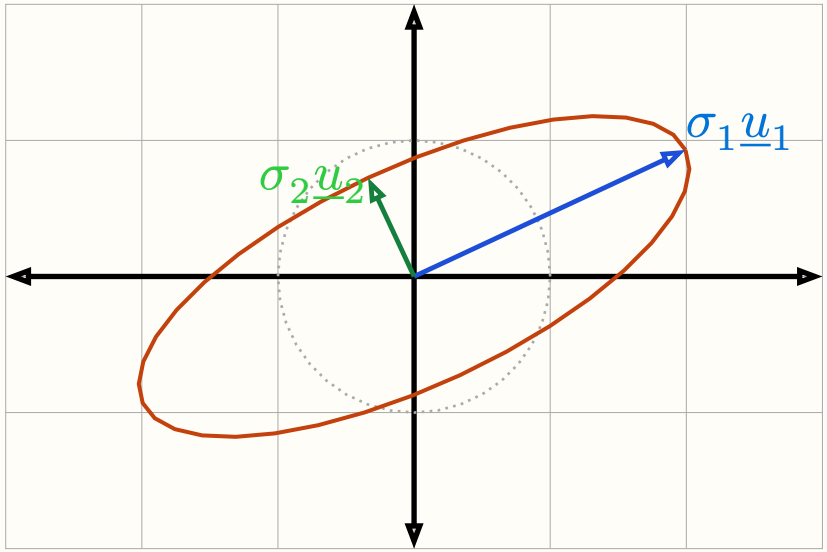
$$S = U\Sigma$$

Eve is unique, unrotated! ($U_{\bullet,2}$)

Two "*personalities*" react to separate "*movie types*"

$$A = U\Sigma V^T$$

Every matrix has a Singular Value Decomposition (SVD)



Unit sphere:

$$S = \{\underline{v} : \|\underline{v}\|_2 = 1\}$$

Transformed sphere (ellipse):

$$R = \{A\underline{v} : \|\underline{v}\|_2 = 1\}$$

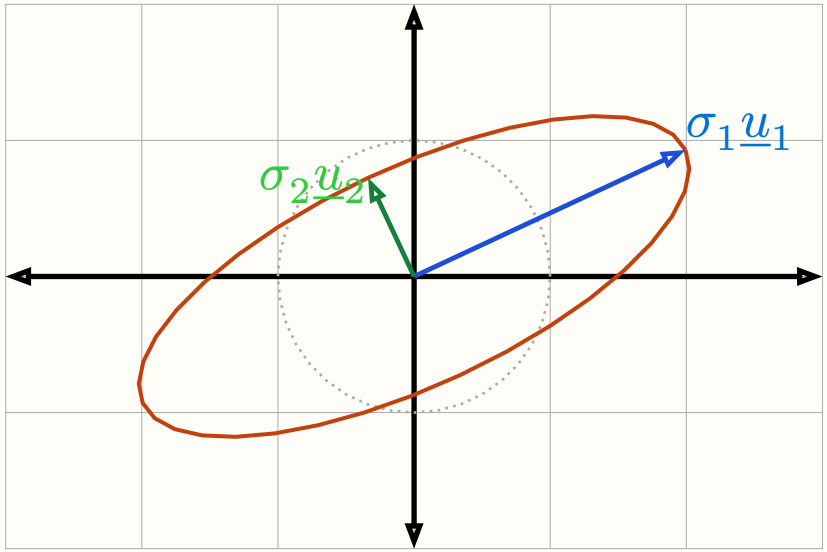
Works for any matrix

$$A = U\Sigma V^T$$
$$= \text{rotation} \circ \text{scaling} \circ \text{rotation}$$

Netflix: $\Sigma = \begin{bmatrix} 2\sqrt{103} & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

- Two singular values σ_i

Every matrix has a Singular Value Decomposition (SVD)



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Transformed sphere (ellipse):

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Questions

- Construct diagonal matrix Σ. Why is calculating Σ^2 easy?
- If $A = U\Sigma V^T$, what is A^2? A^n?
- SVD the grocery matrix and interpret output